

LESSON 2-7 • HOMEWORK PACKET • SHEET 1 OF 2

Dividing a Decimal by a Whole Number

Maryland Standard 6.NS.B.3 • Learning target – I can divide a decimal by a whole number and place the decimal point correctly.

REMEMBER

- Divide as you do with whole numbers. Put the decimal point in the quotient **straight above** the point in the dividend.
- If an amount is left over, **annex a zero** (write 0 after the last digit) and keep dividing.
- **Check:** quotient $\times$ divisor = dividend. Estimate first to catch a misplaced point.

WORKED EXAMPLE

Four friends share 22.2 kg of potting soil equally. How much does each get?

Step 1 Estimate: $22 \div 4$ is about 5.5. Put the point straight above.

Step 2 $22 \div 4 = 5$ R2. Then 22 tenths $\div 4 = 5$ tenths R2.

Step 3 Annex a zero: 20 hundredths $\div 4 = 5$ hundredths.

Answer **5.55 kg each ($5.55 \times 4 = 22.2$)**

PRACTICE*show your work • label units*

1. Find each quotient.

a. $9.6 \div 3 =$ _____

b. $18.5 \div 5 =$ _____

c. $2.94 \div 7 =$ _____

d. $0.56 \div 8 =$ _____

e. $31.2 \div 6 =$ _____

f. $9.3 \div 6 =$ _____

2. **Bulk Rice.** A 12-pound bag of rice costs \$16.80. What is the cost per pound?

Show your work.

Answer _____ per pound

3. **Relay Race.** 8 runners split a 10.2-kilometer relay equally. How far does each runner go?

Show your work.

Answer _____ km

4. **Better Buy.** Store A sells 4 cans of beans for \$3.56. Store B sells 6 cans for \$5.10.

a. A, per can: \$ _____

b. B, per can: \$ _____

c. Better buy: _____

5. **Find the Error.** Marcus wrote $4.35 \div 5 = 8.7$. Use an estimate to explain his mistake. Then give the correct quotient.

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about # _____

AT HOME At the store, find a pack of several items. Divide the price by the number of items to find the cost of one.

LESSON 2-7 • HOMEWORK PACKET • SHEET 2 OF 2**Dividing by a Decimal**

Maryland Standard 6.NS.B.3 · Learning target – I can divide by a decimal by making the divisor a whole number first.

REMEMBER

- Make the divisor whole: move **both** decimal points the **same** number of places right. The quotient does not change: $4.8 \div 0.6 = 48 \div 6$.
- Annex zeros in the dividend if needed. Dividing by a number less than 1 gives a quotient **greater** than the dividend.

WORKED EXAMPLE

How many 0.15-liter cups can be filled from 4.5 liters of juice?

Step 1 Make 0.15 whole: $0.15 \times 100 = 15$.**Step 2** Do the same to 4.5: $4.5 \times 100 = 450$.**Step 3** $450 \div 15 = 30$. Check: $30 \times 0.15 = 4.5$ **Answer** 30 cups**PRACTICE***show your work · label units*

1. Rewrite with a whole-number divisor. Then divide.

a. $4.8 \div 0.6 = \underline{\hspace{2cm}}$

b. $2.75 \div 0.5 = \underline{\hspace{2cm}}$

c. $0.96 \div 0.12 = \underline{\hspace{2cm}}$

d. $9 \div 0.3 = \underline{\hspace{2cm}}$

e. $1.68 \div 0.04 = \underline{\hspace{2cm}}$

f. $13.2 \div 1.1 = \underline{\hspace{2cm}}$

- 2.
- Pillows.**
- Leon has 7.5 yards of fabric. Each pillow uses 1.25 yards. How many pillows can he make?

*Show your work.***Answer** _____ pillows

- 3.
- Juice Glasses.**
- A pitcher holds 2.4 liters of juice. How many glasses can it fill?

a. 0.3-liter glasses: _____

b. 0.2-liter glasses: _____

- 4.
- Walking Path.**
- One lap of a park path is 0.75 mile. How many laps make each distance?

a. 4.5 miles: _____ laps

b. 6 miles: _____ laps

- 5.
- Find the Error.**
- Keisha wrote
- $2.4 \div 0.08 = 24 \div 8 = 3$
- . Explain her error. Find the correct quotient.

TEST PRACTICE*choose the best answer*

6. Which division has the same quotient as
- $6.72 \div 0.8$
- ?

A $672 \div 8$

B $67.2 \div 8$

C $6.72 \div 8$

D $67.2 \div 80$

7. How many 0.4-pound bags can be filled from 15.6 pounds of trail mix?

A 3.9

B 6.24

C 39

D 390

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about # _____**AT HOME** Tell someone at home why $6 \div 0.5$ is greater than 6.